

6.1 The laws of indices

- First law

$$a^m \times a^n = a^{m+n}$$

- Second law

$$\frac{a^m}{a^n} = a^{m-n}$$

- Key point: any number raised to the power 0 equals 1, that is $a^0 = 1$

- Third law

$$(a^m)^n = a^{mn}$$

- Key point: $(a^m b^n)^k = a^{mk} b^{nk}$

6.2 Negative powers

- key points - $a^{-m} = \frac{1}{a^m}$, $a^m = \frac{1}{a^{-m}}$

6.3 Square roots, cube roots and fractional powers

- Key point: $x^{\frac{1}{2}} = \sqrt{x}$, $x^{\frac{1}{3}} = \sqrt[3]{x}$
- Key point: $x^{\frac{1}{n}} = \sqrt[n]{x}$
- Key point: $a^{\frac{m}{n}} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$